

1. Explain the importance of the OSI model in network communication.
2. Describe all seven layers of the OSI model with their functions.
3. Compare the OSI and TCP/IP models.
4. Explain the layers of the TCP/IP model with examples.
5. Differentiate between the OSI model and TCP/IP model.
6. Differentiate between Class A, B, C, D, and E IP addresses.
7. Explain public and private IP addresses with examples.

Numerical Questions:

8. Identify the class of the following IP addresses:
 - 10.20.30.40
 - 172.16.1.1
 - 192.168.10.5
9. Find the network ID and host ID of 192.168.1.100/24.

Short Questions:

1. What is a subnet mask?
2. Why is a subnet mask used?
3. What is the default subnet mask of Class C?
4. What does /24 represent?
5. What is the subnet mask for /16?

Long Questions:

1. Explain the role of subnet masks in IP networking.
2. How does a subnet mask determine the network and host portions?
3. Differentiate between classful and classless subnet masks.

Numerical Questions:

1. Subnet the network 192.168.1.0/24 into 4 equal subnets.
2. Find Network Address, Broadcast Address, and Valid Host Range for 192.168.10.34/27.
3. How many hosts can be accommodated in a /26 subnet?

7. FLSM, VLSM and CIDR

Short Questions:

1. What is FLSM?

2. What is VLSM?
3. What is CIDR?
4. Which is more efficient: FLSM or VLSM?
5. What does /30 mean in CIDR notation?

Long Questions:

1. Differentiate between FLSM and VLSM.
2. Explain CIDR and its advantages.
3. Describe how VLSM helps in efficient IP address allocation.
4. Compare FLSM, VLSM, and CIDR.

Numerical Questions:

1. Divide 192.168.1.0/24 into 100, 50, and 25 hosts using VLSM.
2. Using FLSM, create 8 equal subnets from 172.16.0.0/16.
3. Determine the network, broadcast, and host range for 10.10.10.65/26.